Knowing What You Are Owed: A Retrieval-Augmented Chatbot for Philippine Consumer-Rights Questions on Online Purchases

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I. Selected UN SDG and Justification

Selected SDG. SDG 12 – Responsible Consumption and Production.

Relevant SDG area. Target 12.8 [1], which seeks to ensure that people everywhere have the relevant information and awareness for sustainable lifestyles, with a direct operational link to Target 12.5 on reducing waste generation through prevention, reduction and reuse.

Justification. The connection between consumer rights and SDG 12 is not decorative; it is mechanical. Republic Act No. 7394, the Consumer Act of the Philippines [2], grants a buyer three remedies for a defective product, in a fixed order: repair, replacement, and refund. That ordering is the waste hierarchy expressed as a legal entitlement. A consumer who knows a seller is obliged to repair a defective item keeps the item. A consumer who does not know simply discards it and buys another, and a product that could have been repaired becomes waste. The information gap therefore has a physical output measured in discarded goods, which is precisely what Target 12.5 addresses and precisely what Target 12.8 identifies information as the lever for.

Connection to RAG. This domain carries an unusually sharp hallucination risk [3], which is what makes it suitable for a grounded system. A general-purpose language model trained largely on English-language text has absorbed a great deal of American and European consumer law – the European Union's fourteen-day right of withdrawal, for instance, which allows a buyer to return goods simply for changing their mind. No such right exists under Philippine law. A model answering from pre-training will supply that rule fluently and confidently, and the answer will be wrong in a way the user cannot detect. A retrieval-augmented chatbot [4] restricted to Philippine statutes and Department of Trade and Industry issuances, and required to refuse when those documents are silent, is the appropriate design for a domain where the most plausible-sounding wrong answer comes from another jurisdiction.

II. Narrow Problem Statement

TABLE I. Narrowing from Broad Domain to Specific Information Problem

Level	Group output
Broad domain	Consumer protection and responsible consumption
SDG-related issue	SDG 12, Target 12.8: access to the information a consumer needs, linked to Target 12.5 through repair and reuse
Target users	Filipino online shoppers, particularly students buying on e-commerce platforms
Narrow problem	Buyers cannot tell which remedies they are legally entitled to when an online purchase arrives defective, because sellers state policies that are not the law and platforms state policies that are not the law either
Proposed chatbot focus	Short questions with a statutory answer: whether a stated policy is lawful, which of the three remedies applies, what does not qualify, and how a complaint is filed

Problem statement. The users are Filipino online shoppers, particularly students, who buy from e-commerce platforms and occasionally receive goods that are defective or not as described. Their difficulty is that the rules governing their situation are stated by three parties who do not agree: the seller, who may post a policy the law forbids; the platform, whose internal refund window is a commercial term rather than a legal right; and the statute, which almost no buyer has read. A shopper deciding whether to pursue a remedy is therefore reasoning from the least authoritative of the three sources. The problem is worth investigating because the statutory answers are crisp and checkable, because the documents are public and require no permission, and because a wrong answer has a measurable cost: a buyer who believes nothing can be done keeps a defective item, discards it, or buys a replacement, which is the waste outcome Target 12.5 exists to prevent.

III. Primary Research Question

To what extent does retrieval grounding improve the correctness and groundedness of answers to Philippine consumer-rights questions about online purchases, compared with a non-RAG baseline using the same language model?

Justification. The question is appropriate because it interrogates the claim the design rests on – that retrieval, not model capability, is what keeps answers within the correct jurisdiction – rather than asking whether a chatbot can be built. It is focused because it fixes the domain, the corpus described in Section IV, and the comparison, leaving retrieval as the only free variable. It is measurable on two axes that can be scored against evidence: correctness, by comparing each answer with the statutory passage established as its ground truth, and groundedness, by checking whether every claim in an answer is supported by a retrieved passage. Holding the language model constant across both arms is what makes any observed difference attributable to retrieval rather than to the model.

Why this domain sharpens the comparison. Jurisdictional error is a failure mode that can be identified without ambiguity. If the baseline supplies a cooling-off period, a fourteen-day return window, or a small-claims threshold drawn from foreign law, that is a scoreable hallucination rather than a matter of interpretation. The domain therefore supplies a clean signal for the very effect the

question asks about.

IV. Initial Knowledge-Base Data Sources

All sources below are public and require no permission to use. None has been collected or verified yet; collection belongs to the implementation phase.

A. Republic Act No. 7394, the Consumer Act of the Philippines [2]. *Document type:* statute. *Relevance:* the governing law, including the prohibition on deceptive sales acts and the repair, replacement and refund remedies. *Availability:* public full text. *Reliability and permission:* primary legal source in the public domain.

B. DTI Fair Trade Enforcement Bureau consumer FAQ [5]. *Document type:* official frequently asked questions. *Relevance:* the agency's plain-language reading of the statute, including its position that a posted "No Return, No Exchange" policy is prohibited. *Availability:* public webpage. *Reliability and permission:* the implementing agency; content in the public domain.

C. DTI consumer advisories and information materials. *Document type:* advisory and brochure. *Relevance:* states the remedies and their limits in the form a consumer actually encounters. *Availability:* public download. *Reliability and permission:* official agency publication.

D. DTI consumer complaint procedure and mediation rules. *Document type:* procedural guidance and department administrative orders. *Relevance:* supplies the escalation path: where a complaint is filed, what it must contain, that no filing fee is charged, and that the matter proceeds to a mediation conference. *Availability:* public. *Reliability and permission:* the implementing agency.

E. Republic Act No. 8792, the Electronic Commerce Act [6]. *Document type:* statute. *Relevance:* establishes the legal standing of electronic transactions and documents, which is what connects a 1992 consumer statute to a purchase made on a present-day e-commerce platform. *Availability:* public full text. *Reliability and permission:* primary legal source in the public domain.

V. Initial Evaluation Data Plan

Intended source of questions. The questions will be collected rather than invented, from three places: a short written survey of classmates who shop online, questions already visible in group chats when a delivery goes wrong, and the question categories the agency itself reports as most frequently asked. The survey has not yet been conducted.

Types of questions. The set will include legality checks on a stated policy, remedy-selection questions, boundary questions about what does not qualify, procedural questions about filing and cost, evidentiary questions about what records are needed, and a deliberately unanswerable category.

Size of the test set. The target is thirty to forty questions, with roughly one in five drawn from the unanswerable category. That range is deliberate on both sides: large enough that a single item cannot move the result, and small enough that every ground-truth mapping can be verified by hand rather than assumed. A set this size also keeps the proportion of unanswerable items meaningful, since a false-refusal or false-answer rate computed over a handful of questions would indicate a failure mode without establishing a rate.

Establishing expected answers. Each test question will be mapped to the specific supporting passage – statute article or agency issuance – that contains its answer. Two group members will perform this mapping independently and the third will reconcile disagreements, so that the ground truth is not one person's reading. A question for which no supporting passage can be located is either removed or reclassified as unanswerable.

Including unanswerable questions. This domain supplies a natural and meaningful unanswerable category: questions about *platform policy* rather than law. An individual marketplace's refund window, its internal dispute timeline, or a particular seller's shipping terms are commercial arrangements that appear nowhere in the statute. They measure whether the system correctly declines rather than inventing a legal rule to fill the gap, and the distinction they test, between what a platform promises and what the law requires, is the same distinction the users themselves are failing to make.

VI. Possible Language Model and Justification

Model name. openai/gpt-oss-20b, accessed through the Groq inference API.

Justification. The model is proposed on four grounds. It is available on a free API tier, so the study requires no local GPU and no budget. It is small enough to answer quickly. It follows system-prompt constraints reliably, which is the property the grounding contract depends on, since a model that ignores an instruction to answer only from context cannot be evaluated for groundedness. And it has been exercised in the course laboratory work, so its behavior with a retrieval chain is known rather than assumed.

Expected role. The model is a reader and synthesizer, not a source of legal knowledge. Retrieved passages are inserted into a context placeholder in the system prompt, and the model is instructed to answer strictly from them and to emit a fixed refusal sentence when they are insufficient. Any legal rule appearing in an answer that is absent from the retrieved passages counts as a failure, even if that rule is true somewhere else in the world.

Practical considerations. Access is through a free tier with rate limits adequate for a student evaluation, and the key is supplied at runtime rather than stored in the notebook. Deployment is hosted, so no local hardware is required. Hosted models are deprecated over time, and the model named in the course template has already been retired, so the implementation should resolve an available model at run time rather than hard-coding one. The corpus is in English while users frequently ask in Filipino or code-switched Taglish, which is a known limitation to be tested rather than assumed away. This is a preliminary proposal: no claim is made that this model is the best choice, since establishing that would require a comparison the present study does not run.

VII. Group Reflection

Why is this problem sufficiently narrow and manageable for a student research project?

The corpus is bounded and fully public – two statutes and a small set of agency issuances – small enough to curate carefully and verify by hand, which is what makes a trustworthy ground truth possible, and obtainable without requesting permission from any institution. The questions are short and have statutory answers, so each maps to a specific passage, avoiding the scoring ambiguity that open-ended questions create. And the evaluation is a comparison rather than an open exploration: the same questions, the same model, with and without retrieval.

It is manageable for a further reason. The scope excludes everything that would make it unmanageable: no legal advice

is offered, no platform is evaluated, no claim is made about actual consumer outcomes, and no jurisdiction outside the Philippines is covered. The study asks one question about one corpus with one model, and answers it with evidence.

With three members the workload divides cleanly into corpus curation, test-set construction, and evaluation scoring, with the ground-truth mapping deliberately duplicated across members so that no single reading determines a result.

REFERENCES

[1] United Nations, *Transforming Our World: The 2030 Agenda for Sustainable Development*, A/RES/70/1, New York, USA, 2015.

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[3] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023.

[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. NeurIPS*, 2020, pp. 9459–9474.

[5] Department of Trade and Industry, "Is 'No Return, No Exchange' policy allowed?" Fair Trade Enforcement Bureau, 2022. fairtrade.dti.gov.ph

[6] Republic Act No. 8792, *Electronic Commerce Act of 2000*, Manila, Philippines, 2000.